COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	07/10/2026	**DATE SUBMITTED:**	07/10/2026
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ACTIVITY NO. 1 SEATWORK

1. PROCEDURE OUTPUT

SECTIONS	ANSWER
Header File Declaration Section	`#include<iostream>` `#include <string>`
Global Declaration Section	N/A

Class Declaration and Method Definition Section	`//Create a function that will take two numbers and display the sum`
	
	`float Sum(float n1, float n2) {`
	`return n1 + n2;`
	`}`
	
	`// Create a function that will return whether variable A is greater than variable B.`
	
	`std::string Compare(float n1, float n2){`
	`if(n1 > n2) {`
	`return "Variable A is Bigger than variable B";`
	`}`
	`else {`
	`return "Variable B is bigger than variable A";`
	`}`
	`}`

	`// Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success`
	
	`std::string CompareBool(int a, int b) {`
	`if ((a != 0 && a != 1) || (b != 0 && b != 1)) {`
	`return "Invalid input! Please enter only 0 or 1.";`
	`}`
	
	`bool valA = (a == 1);`
	`bool valB = (b == 1);`
	
	`if (valA && valB) {`
	`return "Both are true";`
	`} else if (!valA && !valB) {`
	`return "Both are false";`
	`} else {`
	`return "At least one value is true";`
	`}`
	`}`

Main Function	`int main() {`
	
	`float n1, n2;`
	`Int a, b;`
	`std::string result;`
	
	`std::cout << "Enter 1st Number: ";`

```
        std::cin >> n1;
  std::cout << "Ener 2nd Number:
            ";
        std::cin >> n2;

    std::cout << "Enter first
boolean value(1 = True or 0 =
        False): ";
        std::cin >> a;
    std::cout << "Enter seond
boolean value(1 = True or 0 =
        False): ";
        std::cin >> b;

    result = Compare(n1, n2);

   std::cout << "The sum of 2
number is " << Sum(n1, n2) <<
        std::endl;
     std::cout << result <<
        std::endl;
  std::cout << CompareBool(a,
          b);

        return 0;

          }
```

Method Definition	N/A

ILO A: Create Code That Follows the Basic C++ Code Structure

A function prototype in c++ is a declaration of the name, parameters and return type of the function before its definition. Write a C++ code the satisfies the following:

- Create a function that will take two numbers and display the sum.

```cpp
#include<iostream>
#include <string>

//Create a function that will take two numbers and display the sum

float Sum(float n1, float n2) {
    return n1 + n2;
}
```

- Create a function that will return whether variable A is greater than variable B.

```cpp
std::string Compare(float n1, float n2){
    if(n1 > n2) {
        return "Variable A is Bigger than variable B";
        }
    else {
        return "Variable B is bigger than variable A";
        }
}
```

- Create a function that will take two Boolean values and display the result of all logical operations then
- return true if it was a success.

```cpp
// Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success

std::string CompareBool(bool a, bool b){
    if (a && b){
        return "Both is true";
    }
    else if (!a && !b){
        return "Both is false";
    }
    else if (a || b){
        return "Atleast one value is true";
    }
    else {
        return "Wrong value";
    }
}
```

```cpp
int main() {

float n1, n2;
bool a, b;
std::string result;

    std::cout << "Enter 1st Number: ";
    std::cin >> n1;
    std::cout << "Ener 2nd Number: ";
    std::cin >> n2;

    std::cout << "Enter first boolean value(1 = True or 0 = False): ";
    std::cin >> a;
    std::cout << "Enter seond boolean value(1 = True or 0 = False): ";
    std::cin >> b;

    result = Compare(n1, n2);

    std::cout << "The sum of 2 number is " << Sum(n1, n2) << std::endl;
    std::cout << result << std::endl;
    std::cout << CompareBool(a, b);

return 0;

}
```

```
PROBLEMS 3  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\TIPQC\Desktop\ACALA CPE21S2> cd "c:\Users\TIPQC\Desktop\ACALA CPE21S2\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }
Enter 1st Number: 100
Ener 2nd Number: 150
Enter first boolean value(1 = True or 0 = False): 1
Enter seond boolean value(1 = True or 0 = False): 0
The sum of 2 number is 250
Variable B is bigger than variable A
Atleast one value is true
PS C:\Users\TIPQC\Desktop\ACALA CPE21S2>
```

Code Analysis:
On the First part i did a simple function that add 2 float numbers and return those values. On the second one i just use a operator and a if-else statement to compare 2 float values which i also used on the first part. Lastly the boolean part i just use if-else also to state if it's "both true" "both false". Most of the gathered value is in the main.

ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements

SECTIONS	ANSWER
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	`const bool Triangle::validateTriangle() { return (totalAngle <= 180); }`
Class Declaration and Method Definition Section	`class Triangle { private: double totalAngle, angleA, angleB, angleC; public: Triangle(double A, double B, double C);`

	`void setAngles(double A, double B, double C);`
	`const bool validateTriangle();`
	`};`
	
	`Triangle::Triangle(double A, double B, double C) {`
	`angleA = A;`
	`angleB = B;`
	`angleC = C;`
	`totalAngle = A + B + C;`
	`}`
	
	`void Triangle::setAngles(double A, double B, double C) {`
	`angleA = A;`
	`angleB = B;`
	`angleC = C;`
	`totalAngle = A + B + C;`
	`}`

Main Function	`int main() {`
	`Triangle set1(40, 30, 110);`
	`if (set1.validateTriangle()) {`
	`std::cout << "The shape is a valid triangle.\n";`
	`} else {`
	`std::cout << "The shape is NOT a valid triangle.\n";`
	`}`
	`return 0;`
	`}`

Method Definition	`public:`
	`Triangle(double A, double B, double C);`
	`void setAngles(double A, double B, double C);`

```
    const bool validateTriangle();
                };

    Triangle::Triangle(double A,
        double B, double C) {
                angleA = A;
                angleB = B;
                angleC = C;
            totalAngle = A + B + C;
                }

void Triangle::setAngles(double A,
        double B, double C) {
                angleA = A;
                angleB = B;
                angleC = C;
            totalAngle = A + B + C;
                }

            const bool
    Triangle::validateTriangle() {
        return (totalAngle <= 180);
                }
```



Code Analysis:

Using private variables enforces data protection angles can only be changed through methods

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C: Solve Different Problems using the C++ Programming Language.

1. Create a C++ program to swap the two numbers in different variables.

```
temp = a;
a = b;
b = temp;
```

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

```
float KtoF(float K){
    return (K - 273.15) * 9/5 + 32;
}
```

3. Create a C++ program that has a function that will calculate the distance between two points.

```
float Distance(float x1, float y1, float x2, float y2) {
    return std::sqrt(std::pow(x2 - x1, 2) + std::pow(y2 - y1, 2));
}
```

4. Modify the code given in ILO B and add the following functions:
   a. A function to compute for the area of a triangle
   b. A function to compute for the perimeter of a triangle
   c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

SUPPLEMENTARY.cpp | TRIANGLE.cpp | main.cpp

TRIANGLE.cpp > Triangle > triangleType()

```cpp
class Triangle {
    double perimeter(double a, double b, double c) {
        return a + b + c;
    }


    std::string triangleType() {
        if (angleA < 90 && angleB < 90 && angleC < 90) {
            return "Acute-angled triangle";
        } else if (angleA > 90 || angleB > 90 || angleC > 90) {
            return "Obtuse-angled triangle";
        } else if (angleA == 90 || angleB == 90 || angleC == 90) {
            return "Right-angled triangle";
        } else {
            return "Others";
        }
    }
};

int main() {
    Triangle set1(40, 30, 110);

    if (set1.validateTriangle()) {
        std::cout << "The shape is a valid triangle.\n";
        std::cout << "Triangle type: " << set1.triangleType() << std::endl;
        std::cout << "Area (base=10, height=5): " << set1.area(10, 5) << std::endl;
        std::cout << "Perimeter (sides=3,4,5): " << set1.perimeter(3, 4, 5) << std::endl;
    } else {
        std::cout << "The shape is NOT a valid triangle.\n";
    }

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Programming\C++\SCHOOL\CPE 010> cd "c:\Programming\C++\SCHOOL\CPE 010\" ; if ($?) { g++ TRIANGLE.cpp -o TRIANGLE } ; if ($?) { .\TRIANGLE }
The shape is a valid triangle.
Triangle type: Obtuse-angled triangle
Area (base=10, height=5): 25
Perimeter (sides=3,4,5): 12
PS C:\Programming\C++\SCHOOL\CPE 010>
```

SUPPLEMENTARY.cpp

SUPPLEMENTARY.cpp > main()

```cpp
#include <iostream>
#include <cmath>

float KtoF(float K){
    return (K - 273.15) * 9/5 + 32;
}

float Distance(float x1, float y1, float x2, float y2) {
    return std::sqrt(std::pow(x2 - x1, 2) + std::pow(y2 - y1, 2));
}

int main () {
    int a, b, temp;
    float K, x1, x2, y1, y2;
    std::cout << "Enter first number: ";
    std::cin >> a;
    std::cout << "Enter second number: ";
    std::cin >> b;
    std::cout << "Enter temperature in Kelvin: ";
    std::cin >> K;
    std::cout << "Enter coordinates (x1, y1, x2, y2): ";
    std::cin >> x1 >> y1 >> x2 >> y2;

        temp = a;
        a = b;
        b = temp;

    std::cout << "After swapping a = " << a << ", b =  " << b << std::endl;
    std::cout << "Temperature in Fahrenheit: " << KtoF(K) << std::endl;
    std::cout << "Distance between points: " << Distance(x1, y1, x2, y2) << std::endl;

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Programming\C++\SCHOOL\CPE 010> cd "c:\Programming\C++\SCHOOL\CPE 010\" ; if ($?) { g++ SUPPLEMENTARY.cpp -o SUPPLEMENTARY } ; if ($?) { .\SUPPLEMENTARY }
Enter first number: 102
Enter second number: 111
Enter temperature in Kelvin: 41
Enter coordinates (x1, y1, x2, y2): 1 2 5 5
After swapping a = 111, b =  102
Temperature in Fahrenheit: -385.87
Distance between points: 5
```

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Microsoft Copilot**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Example generation, explanation of concept, and asking what is the error**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I just ask for help me or give me tips or what is the error to solve on my own**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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